

Practical 1 guidelines

Print this out and bring it to the practical. You need to complete this during the practical session.

After completing the form, log on to ClickUP and fill in the answers online(Practical 1).

NB: This form will not count any marks, only the answers you submit to ClickUP will count marks. This file is merely here for your convenience - since I give many additional hints and help in this file.

NB: Use the decimal point for number separation. Do not use the comma. eg. 2.1 instead of 2,1. Marks will be lost if the comma is used. Also, do not supply units!

Question 1

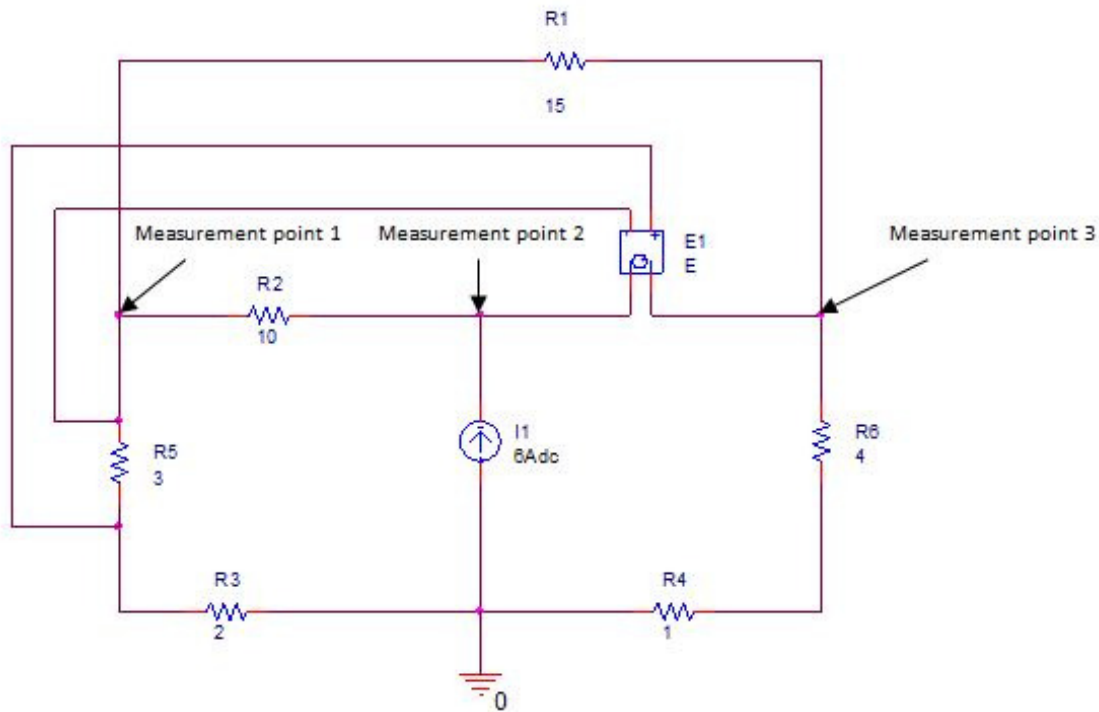


FIGURE 1

Create the above circuit in OrCAD. And then answer the questions that follow.

Hints:



If you struggle to find the DC current source, the part name is **I dc**. Search for it under parts



If you struggle to find the voltage controlled voltage source, search for it by looking for the component named **E**.

Remember that the values underneath the resistors represent their resistances.



Measure voltage at a node

Measure voltage drop across a component

Measure current through a component

Show voltage of all elements after simulation

Show current of all elements after simulation

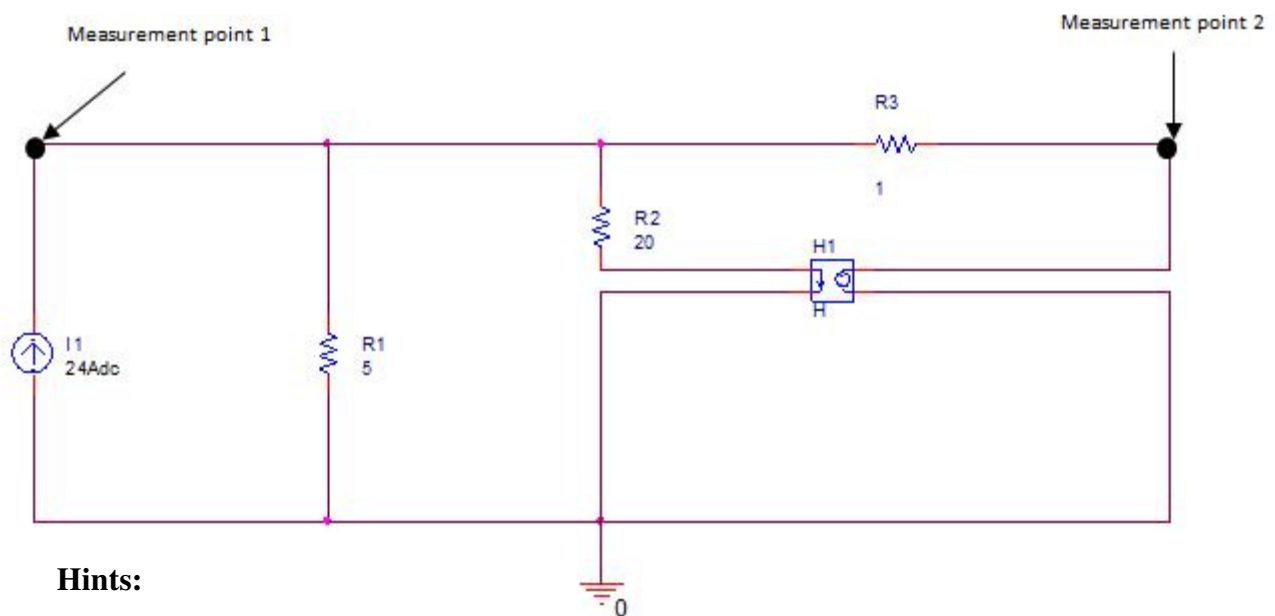
Show power of all elements after simulation

1.1) For the circuit in figure 1, what is the voltage at measurement point 1?

1.2) For the circuit in figure 1, what is the voltage at measurement point 2?

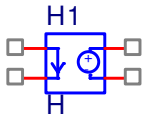
1.3) For the circuit in figure 1, what is the voltage at measurement point 3?

Question 2



Hints:

FIGURE 2



If you struggle to find the current controlled current source, search for it by looking for the component named **H**.

Questions:

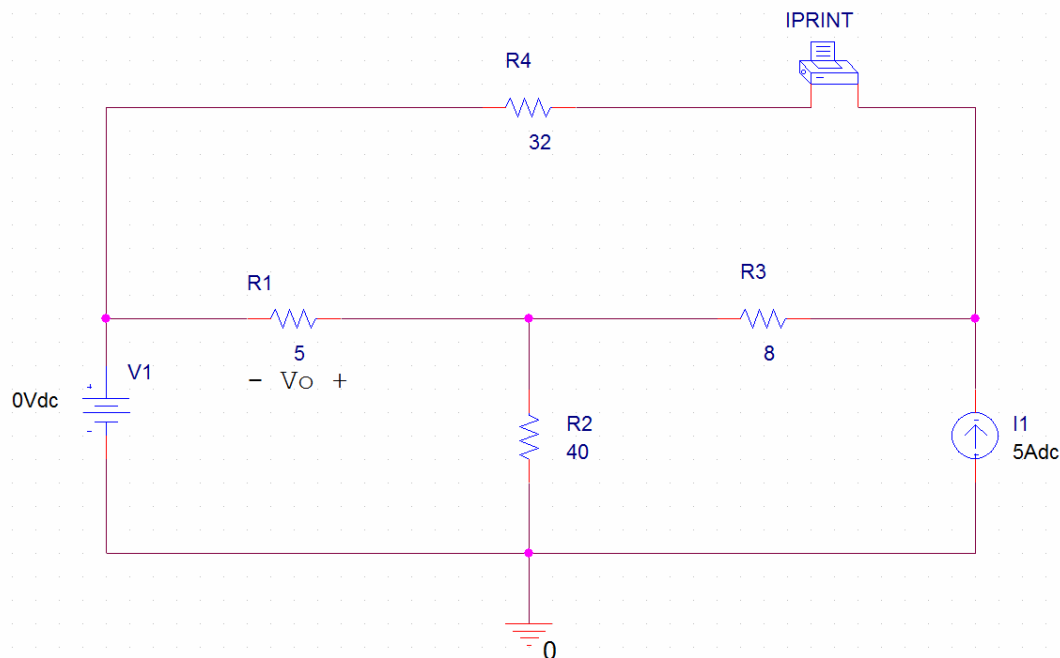
2.1) For the circuit in figure 2, what is the voltage at measurement point 2?

2.2) For the circuit in figure 2, what is the current flowing through the 1 Ohm resistor?

2.3) For the circuit in figure 2, what is the total power dissipated (give the answer in W)?

2.4) For the circuit in figure 2, what is the total power provided (Give the answer in W)?

Question 3.1



Create and simulate the circuit in over the range 0 to 100 V in 20 V

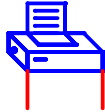
Figure 3

g the "DC Sweep" analysis type. Sweep V1 to 5A. Answer the questions that follow.

Hints:

If you struggle to find the current printer display device, search for it by looking for the component named **IPRINT**.

IPRINT



The current printer is used to display the values of the current flowing through R4. It functions as an Ammeter and effectively 'prints' the values of I_o .

The polarities of the printer determines in which direction it will measure the current. If the polarity is flipped (refer to figure), flip it back by right clicking on the IPRINT device and selecting "mirror horizontally".

To display the values of IPRINT, simulate the circuit. The Schematics window should open with a blank graph. Select Trace - Add Trace.
Click on I(PRINT1). This should now be written in the box 'Trace Expression' at the bottom of the window. Click OK.

Questions:

3.1.1) What is the maximum current that flows through R4?

3.1.2) What is the minimum current flowing through R4?

Hints: To take accurate measurements from the graph, use the Cursor tool. In the schematics window you should see this icon:



Click on it, then the Probe Cursor window should appear from which you can take accurate measurements:

Click on the graph and move the cursor with the keyboard directional.

Probe Cursor			
A1 =	48.933,	1.2777	
A2 =	0.000,	1.4000	
diff=	48.933,	-122.332m	

Questions:

3.1.3) What is the minimum value of the voltage V_o ?

3.1.4) What is the maximum value of the voltage V_o ?

3.1.5) Complete the table below.

Voltage of V1 (Volt)	Voltage of V_o (Volt)
0	
20	
40	
60	
80	
100	

Hints:

Remember to use the correct voltage probes in order to measure voltage across a resistor. Also note the polarity of V_o (refer to figure 3).

Read the values of V_o off of the graph using the cursor tool.

Question 3.2

For the circuit created in question 3.1 (figure 3), set $V1 = 0$ V. Perform another DC sweep, but this time use the current source $I1$ as the sweep variable, and sweep it from 0 to 5 A in 1 A increments.

3.2.1) Complete the table below. Provide answers correct to 2 decimal places.

Current $I1$ (Ampere)	$I(Print1)$ (Ampere)
0	
1	
2	
3	
4	
5	

Now log on to ClickUP and fill in your answers in the:

PRACTICAL 1 ONLINE EVALUATION

Remember to use the decimal point and not the comma for number separation

